Tab 1

	Pimpri Chinchwad Education Trust's Pimpri Chinchwad College of Engineering (PCCoE) (An Autonomous Institute) Affiliated to Savitribai Phule Pune University(SPPU)	

Group No- SE8
Group Members- 123B1D027 Anushka Misal
123B1D034 Yadnesh Patil
123B1D035 Shashwati Pawar

Lab 2: System Requirements and Software Process Model Selection

Title of the Case Study: TourMate – An Intelligent Tourism Management Platform

1. Aim

The aim of this assignment is to analyze the TourMate Tourism Management System and understand how system requirements and development models are selected.
The objectives are:

- Identify the initial system requirements using the MoSCoW prioritization technique.
- Compare different software development process models.
- Select the most suitable model for developing the TourMate system.

2. Problem Statement

The tourism industry involves multiple services such as tour planning, hotel reservations, transportation management, customer handling, and payment processing. Many tourism agencies still depend on manual systems or disconnected software tools, which often leads to errors, slow service, and poor coordination.

TourMate is proposed as a web-based Tourism Management System that helps automate these processes.The system allows customers to explore tour packages, make bookings, complete payments, and provide feedback. Administrators can manage tour packages, track bookings, and generate reports from a single platform.

3. Initial Requirement Identification (MoSCoW Technique)

The MoSCoW technique is used to prioritize system requirements. It categorizes requirements into Must Have, Should Have, Could Have, and Won't Have.

3.1 Must Have Requirements:

- User registration and login
- Secure authentication and role-based access
- Tour package display with destination, price, and duration
- Online booking system
- Booking confirmation
- Admin dashboard to manage tours and bookings
- Secure payment gateway integration
- Database to store users and booking records

3.2 Should Have Requirements:

- Booking history for customers
- Admin approval or modification of bookings
- Hotel accommodation management
- Transport service management
- Basic booking and revenue reports
- Secure password encryption

3.3 Could Have Requirements

- Customer reviews and ratings
- Email or SMS booking notifications
- Personalized tour suggestions
- Customer behavior analysis
- Mobile-friendly interface

3.4 Would Have Requirements

- AI-based travel recommendations
- Integration with airline booking systems
- Virtual tourism using AR/VR
- Multi-language support
- Dynamic pricing using AI

4. Software Process Models

A software process model defines the structured approach used to develop a software system.Selecting an appropriate model helps ensure organized development and successful project completion.

4.1 Model Selected:-

Agile Model:

- Iterative development using short cycles called sprints
- It allows flexible development and continuous improvements.

  Advantages:

  - Flexible requirements
  - Faster feature delivery
  - Early bug detection

  Disadvantages:

  - Requires continuous collaboration

4.2 Agile Development Plan

Sprint 1: User registration, login and browsing tours

Sprint 2: Tour package management

Sprint 3: Booking system

Sprint 4: Payment gateway integration

Sprint 5: Hotel and transport management

Sprint 6: Feedback and reporting system

Sprint 7: Security improvements and optimization

5. Conclusion

TourMate aims to simplify tourism services by automating bookings, payments, and service management. Using MoSCoW prioritization helps identify the most important requirements for the

system. After comparing different software development models, Agile is chosen as the most appropriate model because it supports flexibility, faster development, and continuous improvement.

Tab 2

Assignment 2

System Requirements and Software Process Model Selection

Course: Software Engineering
Case Study: TourMate – Intelligent Tourism Management System

---

1. Aim

The objective of this assignment is to analyze the **TourMate Tourism Management System** and perform the following tasks:

- Identify the **initial broad system requirements** using the **MoSCoW prioritization technique**.
- Study and compare different **software development process models**.
- Select the **most suitable process model** for the development of the system.

---

2. Problem Statement

The tourism industry involves several activities such as managing tour packages, handling customer bookings, coordinating hotel and transport services, processing payments, and collecting feedback from travelers. Many tourism agencies still rely on manual systems or disconnected software tools, which often lead to delays, errors, and poor coordination.

To address these challenges, **TourMate – an Intelligent Tourism Management System** is proposed. TourMate is a web-based system designed to automate tourism services and provide a centralized platform where customers, administrators, and service providers can interact efficiently.

The system will allow users to browse available tour packages, make bookings, process payments, and provide feedback, while administrators will manage tours, monitor bookings, and generate reports.

For successful development of the TourMate system, it is necessary to identify the **initial system requirements** and select an appropriate **software development process model**.

3. Initial Requirement Identification (MoSCoW Technique)

The **MoSCoW method** is used to prioritize requirements into four categories:

- **Must Have** – Essential features required for the system to function
- **Should Have** – Important features that improve the system but are not critical
- **Could Have** – Optional enhancements that improve user experience
- **Won't Have** – Features not included in the current development scope

---

3.1 Must Have Requirements

These requirements are **critical for the basic functioning** of the TourMate system.

- User registration and login functionality
- Secure authentication and role-based access control
- Tour package listing with details such as destination, duration, and price
- Online booking of tour packages
- Booking confirmation generation
- Admin panel for managing tour packages
- Secure payment gateway integration
- Database to store user information, bookings, and transactions

---

3.2 Should Have Requirements

These features **improve system functionality** but are not essential for the first release.

- Booking history for customers
- Admin approval and modification of bookings
- Hotel accommodation management
- Transportation service management
- Basic reporting and analytics for bookings and revenue
- Secure password encryption and session management

3.3 Could Have Requirements

These features provide **additional convenience and improved user experience**.

- Customer reviews and ratings for tour packages
- Email or SMS notifications for booking confirmations
- Personalized tour recommendations
- Advanced data analytics for customer behavior
- Mobile responsive user interface

3.4 Won’t Have Requirements (Current Scope)

These features are **not included in the current system version** but may be considered in the future.

- AI-based travel recommendation system

- Integration with airline reservation systems
- Virtual tourism using AR/VR technology
- Multi-language support
- Dynamic pricing using machine learning

---

4. Software Process Models

Software development process models define the **structure and sequence of activities** involved in developing a software system. Selecting an appropriate model is important to ensure efficient development and project success.

The following models are analyzed for the development of the TourMate system.

---

5. Comparison of Software Process Models

5.1 Waterfall Model

Characteristics

- Linear and sequential development process
- Each phase must be completed before moving to the next phase

Advantages

- Simple and easy to understand
- Clear documentation at every stage

Disadvantages

- Difficult to accommodate changes once development begins
- Testing occurs only after development is completed

Suitability for TourMate

Low suitability because tourism systems often require **frequent updates and flexibility**.

5.2 Incremental Model

Characteristics

- System is developed in **smaller modules or increments**
- Each increment adds new functionality

Advantages

- Early delivery of working features
- Easier testing and debugging
- Allows feedback from users during development

Disadvantages

- Requires careful planning of increments
- System architecture must be well designed from the start

Suitability for TourMate

Moderately suitable because the system can be developed in **modules such as booking, payment, and reporting**.

5.3 Spiral Model

Characteristics

- Combines iterative development with **risk analysis**
- Each cycle includes planning, risk assessment, development, and evaluation

Advantages

- Suitable for large and complex projects
- Strong risk management

Disadvantages

- Expensive and difficult to manage
- Requires experienced teams

Suitability for TourMate

Low suitability for this project because the **system is not large enough to require complex risk management**.

5.4 Agile Model

Characteristics

- Development is done in **short iterations called sprints**
- Continuous feedback from stakeholders

Advantages

- Flexible and adaptable to changes

- Faster delivery of features
- Early detection of errors
- High customer involvement

Disadvantages

- Requires frequent communication among team members
- Documentation may be less detailed compared to traditional models

Suitability for TourMate

Highly suitable because the system may require **continuous updates and improvements**.

---

6. Selected Process Model – Agile

After comparing the available software development models, the **Agile model** is selected as the most appropriate process model for developing the TourMate system.

Agile allows the system to be developed in **multiple iterations**, where each iteration focuses on delivering specific features. This approach makes it easier to test, improve, and update the system based on user feedback.

---

7. Agile Development Plan for TourMate

Sprint	Features Developed
Sprint 1	User registration, login, and tour browsing
Sprint 2	Tour package management
Sprint 3	Online booking system
Sprint 4	Payment gateway integration

Sprint 5	Hotel and transport management
Sprint 6	Feedback system and reporting
Sprint 7	System optimization and security improvements

---

8. Justification for Selecting Agile

The Agile model is suitable for the TourMate system for the following reasons:

- Allows **flexible requirement changes**
- Enables **continuous improvement of system features**
- Supports **faster development cycles**
- Encourages **user feedback and collaboration**
- Helps detect and fix issues early during development

Compared to the Waterfall model, Agile provides **better adaptability for dynamic systems like tourism platforms**.

---

9. Outcome

After completing this assignment, students will be able to:

- Understand how to identify **initial system requirements**
- Apply the **MoSCoW technique for requirement prioritization**
- Compare different **software development process models**
- Select an appropriate model for software system development

---

10. Conclusion

The TourMate Tourism Management System aims to automate tourism-related services such as tour booking, payment processing, and service management. Through the identification of initial requirements using the **MoSCoW technique**, the system's priorities are clearly defined.

After comparing different software development models, the **Agile model** is selected because it offers flexibility, faster delivery, and continuous improvement. This approach ensures efficient development and helps create a reliable and scalable tourism management platform.